

Climate.

- Climate

- Sum total of weather conditions and variations over a large area for a long period of time. (For more than 30 yrs).

Weather

- State of atmosphere over an area at any point of time. (Day to day phenomena).

- Elements of climate and weather remains same.

- It includes →
 - 1) Temperature.
 - 2) Wind.
 - 3) Precipitation.
 - 4) Atmospheric Pressure.
 - 5) Humidity.

- The difference between climate and weather is of time period.

⇒ Climate of India:

- The climate of India has a Monsoon type.

Monsoon → derived from an Arabic word 'Mawsim' which means 'Season'.

In Asia Monsoon type of climate is found in South Asia and South-East Asia.

It includes Bangladesh, Pakistan, Malaysia, Thailand, Vietnam, Singapore, etc.

In Monsoon type of climate there are regional variations. Temperature and precipitation varies from state to state.

e.g. In Summers, Rajasthan → 50°C .

Summer, Kashmir → $20-30^{\circ}\text{C}$.

- In Winter, Dross, Ladakh → -45°C .

- In Winter, Thiruvananthapuram → 25°C .

e.g. Day and Night temperature:-

Rajasthan → Day → 50°C
Night → 15°C } huge difference

- Coastal areas (Kerala) → hardly any difference in temperatures in day and night.

⇒ Precipitation varies in the form, the amount of rainfall and seasonal distribution.

e.g) Upper parts of Him Himalayas → Snowfall
Rest of the country → Rainfall.

e.g) Meghalaya → more than 400cm rainfall
Ladakh → less than 10cm.

e.g) Most of the country → June - Sept. (Rainfall)
Some parts of T.N → Oct - Nov
Kashmir → Winter rainfall.

- Due to these regional variations people have different living, culture, clothing, the food they eat and the kind of house they live in.

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1) Houses in Terai, Goa, Mangalore → have sloping roofs → because of heavy rainfall.

Houses in Rajasthan → Thick walls and ~~thick~~ flat roof.

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To prevent heat from entering into the house

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To preserve water (also provides cooling effect).

⇒ Factors affecting climate (Climatic control)

① Latitude. ② Altitude. ③ Distance from sea.

④ Ocean currents. ⑤ Relief features.

⑥ Pressure and wind system:

① Latitude:

- Temperature decreases from equator towards the poles.

- Temperature at equator is more $\rightarrow$ Tropical or Equatorial type of climate.

- Temperature at poles is less ^{Temperate} and frigid type of climate.

• India $\rightarrow$ tropic of cancer divides it almost into 2 equal halves.

① Part lying south of tropic of cancer $\rightarrow$ Tropical climate.

② Part lying north of tropic of cancer $\rightarrow$ Sub-tropical

• India has both tropical and sub-tropical climate.

② Altitude:- It is height above the sea level.

• As we move up, ~~gases~~ gases become less dense

and temperature decreases.

That's why hilly areas experience milder summers.

eg. Himalayan region and Iran are present at same latitude but different altitudes, thus experience different climates.

Himalayas present in the north prevent cold winds from central Asia to enter into India, thus India experiences milder winters.

③ Distance from the sea: (Continentality)

• Areas present near the sea $\rightarrow$ Moderate Climate.

• Areas present far from the sea $\rightarrow$ Extreme Climate.

$\rightarrow$ It occurs because of land breeze and sea breeze.

e.g. Mumbai, Keralam → Near the sea → Moderate Climate.

Delhi, U.P → far from the sea → Extreme Climate.

④ Ocean Currents.

- It is the continuous movement of water ocean water in a particular direction. It is influenced due to gravity, wind or rotation of earth.

→ Two Types -

1) Warm Ocean Currents. (raise the temperature of place)

2) Cold Ocean Currents. (Drop the temperature of place).

→ El-Nino (Warm Ocean Currents)

La-Nina (Cold Ocean Current).

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cold ocean currents don't have moisture and thus bring little to no rainfall resulting in the formation of deserts in western margins of continents.

⑤ Relief:- Physical features of earth.

• Different relief features have different type of climate.

e.g. Mountain $\begin{cases} \rightarrow \text{Windward side} \rightarrow \text{Rainfall occurs here.} \\ \rightarrow \text{Leeward side} \rightarrow \text{Remain dry.} \end{cases}$

Western Ghat $\begin{cases} \rightarrow \text{West side} \rightarrow \text{Windward side} \rightarrow \text{Rainfall} \\ \rightarrow \text{East side} \rightarrow \text{Leeward} \rightarrow \text{Dry Dry side} \end{cases}$

⑥ Pressure and Wind system :-

- Areas where pressure is high $\rightarrow$ Air sinks, it

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Becomes warmer and drier $\rightarrow$ No rainfall.

$\rightarrow$ Areas where pressure is low $\rightarrow$ Air rises, cools and brings rainfall $\rightarrow$ Moist Climate.

$\Rightarrow$ Winds ~~travel~~ travel from high pressure to low pressure.

$\Rightarrow$ Pressure and Wind System (India):

- 3 wind conditions that influence the climate of India:-

1) Pressure and Surface Wind.

2) Jet streams (Upper air circulation)

3) Western disturbances disturbances.

1) Pressure and Surface Winds:-

Winter

Summer

- | | |
|--|---|
| <ul style="list-style-type: none"> • High pressure is created over north of Himalayas and cold winds blow from this region to low pressure area over Indian ocean to south. • They move over land, thus don't carry any rainfall/moisture. | <ul style="list-style-type: none"> • High pressure develops over Indian ocean and low pressure over north of Himalayas thus winds move from ocean to the mainland (Monsoon). • These winds carry moisture and brings rainfall in the India. |
|--|---|

2) Jet Streams →

- ⇒ • High speed winds that blow above the surface of the earth (10-15 km).
- Speed - 110 km/hr - 184 km/hr.

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- These winds influence monsoon winds and western disturbances

3) Western Disturbances:

- They are the winds that originate from Mediterranean sea and move eastwards towards N and N-W India.
- Bring rainfall and snowfall in the N- and N-W India.

⇒ Features that influence monsoon winds:-

1) Different heating and cooling of land and water:-

- Land and sea heat and cool at different rates

Land → Low Pressure.

Water → High Pressure.

Monsoon winds travel from high pressure to

low pressure.

2) Inter tropical Convergence Zone: (ITCZ):

- It is a low pressure belt near the equator where N-E winds and S-E winds meet.
- During Summer ITCZ shift towards north from equator to India. This will attract the Monsoon winds from high pressure created over sea to low pressure over the land.

3) Presence of high pressure over Madagascar:

- There are low sun rays during summer in southern Hemisphere which creates high pressure over Madagascar and helps monsoon winds to reach India.

4) Intensely heated Tibetan Plateau:

- During summer Tibetan plateau gets heated which forms a low pressure belt over it.
- This attracts monsoon winds and help bring monsoon to India.

⇒ Seasons in India

- There are 4 seasons in India

- 1) The cold weather season (Winter)
- 2) The hot weather season (Summer)
- 3) The advancing monsoon (Rainy season)
- 4) The retreating monsoon (Transition season)

1) Cold Weather Season

- It starts from mid november to february.
- December and january are the coldest months of north India.
- Temperature decreases from south to North.
eg In Chennai temperature is $24-25^{\circ}\text{C}$ while as in Northern plains temperature is $10-15^{\circ}\text{C}$.
- Frost is common and slopes of himalaya experience snowfall.
- N-E winds blow during winters, dry winds that blow on land and don't bring any moisture, thus no rainfall.
- These winds bring some amount of rainfall as

they move over sea (Bay of Bengal) and bring rainfall in some parts of Tamil Nadu.

- During Winter - Western cyclonic disturbances also occur as they bring rainfall and snowfall in North and N-W India.

• Total amount of rainfall during winter is locally c/a Mahavast.

- It is important for growth of rabi crops.

- In peninsular India, $\rightarrow$ no well defined Winter season because the temp. remains moderate because of the distance from the sea.

2) The Hot Weather Season (Summer)

Between - March - May.

- Sun shifts northwards, temp. in north hemisphere increases and pressure falls.
- March → 38°C , April - 42°C , May → 45°C .
- By the end of May, low pressure belt is created over the north India
- During Summer 'Loo' is blown in the northern India.
 - ↓
 - Strong, gusty, dry, hot winds and blow day → They are fatal.
- The season is also characterised by thunderstorm, torrential rains (heavy, rainfall for small period of time), followed by hailstorm.